

Comparative Analysis of Machine Learning and Deep Learning Models for Multi-Level Anomaly Detection in High-Volume Financial Transactions

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Abstract

Financial fraud continues to drain billions from the global economy every year, and the problem grows more complex as digital transactions increase in volume and velocity. This project proposes an enhanced two-stage, multi-level anomaly detection pipeline for financial transactions. The core objective is an empirical head-to-head comparison of five detection models: three from the machine learning domain (Random Forest, Isolation Forest, and XGBoost) and two from deep learning (Autoencoder and LSTM). The pipeline is structured as a hierarchical two-stage framework: Level 1 performs binary anomaly detection (Normal vs. Anomalous), and Level 2 performs risk stratification of flagged transactions into Low, Medium, and High-risk categories. A specific focus is placed on handling extreme class imbalance through SMOTE and evaluating models across both stages using Precision, Recall, F1-Score, and AUPRC.

Keywords—fraud detection, anomaly detection, machine learning, deep learning, SMOTE, class imbalance, hierarchical classification, financial transactions

I. INTRODUCTION

Financial fraud detection represents one of the most challenging problems in applied machine learning: the data is structurally imbalanced, the patterns evolve rapidly, and the cost of both false negatives (missed fraud) and false positives (wrongly flagged legitimate transactions) is high. Financial fraud continues to drain billions from the global economy every year, and the problem grows more complex as digital transactions increase in volume and velocity.

Traditional rule-based systems are fundamentally reactive: they catch patterns seen before while remaining blind to zero-day attacks where fraudsters deploy entirely novel techniques. This creates a double-edged problem: sophisticated fraud slips through undetected, while innocent users are flagged in large numbers, creating a massive operational burden for security teams who must manually review thousands of false alarms.

This project addresses both failure modes by building a two-stage classification pipeline that not only detects anomalies more accurately but also prioritizes them by risk level, enabling financial institutions to allocate investigative resources more efficiently. The specific goals of this revised project are:

- (1) To design and implement a two-stage, multi-level classification pipeline for fraud detection in highly imbalanced financial transaction datasets.
- (2) To conduct a comparative experiment evaluating five models: Random Forest, Isolation Forest, and XGBoost (ML) against Autoencoder and LSTM (Deep Learning).
- (3) To implement Level 1 binary classification (Normal vs. Anomalous) and Level 2 risk stratification (Low / Medium / High Risk) across all applicable models.
- (4) To analyze trade-offs between computational efficiency, detection accuracy, and risk scoring quality to identify the optimal model for real-time financial security applications.
- (5) To produce a fully documented Python implementation and a publishable research paper suitable for conference submission.

II. LITERATURE REVIEW

The existing body of research on financial fraud detection reflects a tension between model interpretability and predictive power. This project builds on that literature by introducing a hierarchical classification objective.

A. Supervised Ensemble Methods

Random Forest [1] remains a widely cited baseline for tabular fraud detection due to its robustness and ability to handle high-dimensional data. XGBoost [4], introduced as an optimized gradient boosting framework, has consistently outperformed Random Forest on structured datasets in benchmark comparisons. Its native handling of class imbalance via the `scale_pos_weight` parameter and its interpretable feature importance scores make it particularly valuable in the financial domain.

B. Unsupervised Anomaly Detection

Isolation Forest [3] offers a fundamentally different approach: rather than learning what fraud looks like, it isolates anomalies by randomly partitioning the feature space. Transactions that require fewer partitions to isolate are classified as anomalies. This unsupervised approach is well-suited to detecting zero-day fraud patterns that supervised models may miss entirely [6].

C. Deep Learning Approaches

Autoencoders [7] address the detection problem by learning a compressed latent representation of normal transaction patterns. When an unseen transaction produces a high reconstruction error, it is flagged as anomalous. LSTM networks [5] offer a complementary strength: by treating transaction histories as temporal sequences, they can detect behavioral anomalies that are invisible in single-transaction snapshots.

D. Handling Class Imbalance

The literature consistently establishes that class imbalance is one of the most critical challenges in fraud detection [2][6]. With fraud rates below 1%, a naive model achieves high accuracy by predicting every transaction as legitimate. SMOTE (Synthetic Minority Over-sampling Technique) addresses this by generating synthetic fraud examples in feature space, enabling models to learn meaningful decision boundaries.

E. Multi-Level Classification

Recent work in fraud risk management has moved beyond binary detection toward risk-tiered frameworks that assign severity scores to flagged transactions [8]. This allows security operations centers to triage alerts by priority rather than treating all flagged transactions equally. This project contributes to this direction by engineering a hierarchical two-stage pipeline on top of each model's output.

III. PROPOSED METHODOLOGY

This project follows the Application-type research path, structured into three main phases.

A. Phase 1: Data Preparation and Feature Engineering

The primary dataset is the Credit Card Fraud Detection Dataset from Kaggle (284,807 transactions; 0.17% fraud rate). Preprocessing includes standard scaling of the Amount and Time features alongside the existing PCA-transformed features (V1-V28). SMOTE will be applied to the training split to address class imbalance for supervised models. For the LSTM pipeline, transactions will be grouped into time-window sequences to capture temporal behavioral patterns.

Multi-level labels will be engineered as follows: Level 1 retains the original binary label (0 = Normal, 1 = Anomalous). Level 2 assigns risk tiers to flagged transactions based on model confidence scores: transactions with low anomaly scores are labeled Low Risk, moderate scores as Medium Risk, and high-confidence anomalies as High Risk.

B. Phase 2: Model Implementation

Five models will be implemented and evaluated: (1) Random Forest — supervised ensemble baseline with Level 2 risk tiers derived from class probability output; (2) XGBoost — gradient boosting classifier with native imbalance handling via scale_pos_weight; (3) Isolation Forest — unsupervised anomaly detection with Level 2 risk tiers derived from anomaly score distribution; (4) Autoencoder (TensorFlow/Keras) — unsupervised with Level 2 tiers derived from reconstruction error thresholds; (5) LSTM (TensorFlow/Keras) — sequence-based supervised model capturing temporal transaction patterns.

C. Phase 3: Evaluation and Comparative Analysis

Since standard accuracy is a misleading metric under class imbalance, performance will be evaluated using Precision, Recall, F1-Score, and the Area Under the Precision-Recall Curve (AUPRC) at Level 1. At Level 2, evaluation will use Macro and Weighted F1-Score, per-class confusion matrices, and Cohen's Kappa. Computational efficiency (training time, inference latency) will also be recorded for each model to assess real-time deployment suitability.

IV. QUARTERLY PLAN

TABLE I
Revised Quarterly Plan

Quarter	Focus Areas	Key Deliverable
Q1 (Month 1)	Extended literature review; EDA; SMOTE preprocessing; multi-level label engineering; LSTM sequence construction	Clean dataset ready for modeling
Q2 (Month 2)	Implement all 5 models; Level 1 binary classification baselines; initial Autoencoder and LSTM training	Working implementations of all models
Q3 (Month 3)	Level 2 risk stratification; hyperparameter tuning; full comparative analysis; visualizations	Final paper and presentation

V. CONCLUSION

This research proposal extends the original scope by adding two high-value model contributions (XGBoost and LSTM) and a hierarchical two-stage classification framework that moves beyond binary fraud detection to risk-tiered anomaly scoring. The result is a more complete and practically relevant pipeline that mirrors how financial institutions triage fraud alerts. By comparing five models across both levels of classification and measuring computational efficiency alongside detection performance, this project aims to provide clear, empirical guidance on which approach offers the best balance of accuracy, interpretability, and operational viability for real-world deployment.

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